26. ADC – Analog to Digital Converter

26.1 Features

- **10-bit Resolution**
- **1 LSB Integral Non-linearity**
- **±2 LSB Absolute Accuracy**
- **13µs - 260µs Conversion Time**
- **Up to 76.9kSPS (Up to 15kSPS at Maximum Resolution)**
- **16 Multiplexed Single Ended Input Channels**
- **14 Differential input channels**
- **4 Differential Input Channels with Optional Gain of 10× and 200×**
- **Optional Left Adjustment for ADC Result Readout**
- **0V - V_{CC} ADC Input Voltage Range**
- **2.7V - V_{CC} Differential ADC Voltage Range**
- **Selectable 2.56V or 1.1V ADC Reference Voltage**
- **Free Running or Single Conversion Mode**
- **Interrupt on ADC Conversion Complete**
- **Sleep Mode Noise Canceler**

The ATmega640/1280/1281/2560/2561 features a 10-bit successive approximation ADC. The ADC is connected to an 8/16-channel Analog Multiplexer which allows eight/sixteen single-ended voltage inputs constructed from the pins of Port F and Port K. The single-ended voltage inputs refer to 0V (GND).

The device also supports 16/32 differential voltage input combinations. Four of the differential inputs (ADC1 & ADC0, ADC3 & ADC2, ADC9 & ADC8 and ADC11 & ADC10) are equipped with a programmable gain stage, providing amplification steps of 0 dB (1×), 20 dB (10×) or 46 dB (200×) on the differential input voltage before the ADC conversion. The 16 channels are split in two sections of 8 channels where in each section seven differential analog input channels share a common negative terminal (ADC1/ADC9), while any other ADC input in that section can be selected as the positive input terminal. If 1× or 10× gain is used, 8 bit resolution can be expected. If 200× gain is used, 7 bit resolution can be expected.

The ADC contains a Sample and Hold circuit which ensures that the input voltage to the ADC is held at a constant level during conversion. A block diagram of the ADC is shown in Figure 26-1 on page 269.

The ADC has a separate analog supply voltage pin, AVCC. AV_{CC} must not differ more than ±0.3V from V_{CC}. See the paragraph “ADC Noise Canceler” on page 275 on how to connect this pin.

Internal reference voltages of nominally 1.1V, 2.56V or AVCC are provided On-chip. The voltage reference may be externally decoupled at the AREF pin by a capacitor for better noise performance.

The Power Reduction ADC bit, PRADC, in “PRR0 – Power Reduction Register 0” on page 55 must be disabled by writing a logical zero to enable the ADC.